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Role: Computer Vision

ADVANCED 2D VIRTUAL TRY-ON

Today, we focused on enhancing the accuracy and realism of the Virtual Try-On system by improving garment fitting, body alignment, and cloth overlay techniques. In the previous implementation, the system mainly performed a basic 2D cloth overlay after body detection and pose estimation. Although the output was functional, the garment placement was not fully aligned with the body posture and sometimes appeared unrealistic due to fixed resizing and direct overlay methods.

In today's work, we improved the virtual try-on pipeline by introducing advanced body alignment techniques using pose estimation and body landmark detection. Special focus was given to shoulder positioning so that the garment aligns naturally with the user's body. Dynamic scaling mechanisms were also implemented, allowing the clothing to automatically resize according to body dimensions instead of using fixed manual scaling.

To improve the realism of the output, transparent PNG garments were used instead of normal clothing images. This helped the clothing blend naturally with the user image. Alpha blending techniques were applied to create smooth cloth integration, reducing the pasted-image appearance seen in the previous task. Cloth segmentation was also integrated to identify garment regions more accurately and intelligently determine the correct overlay area based on body posture and landmarks.

Multiple images with different body positions and poses were tested to validate the performance and consistency of the system. High-resolution front-facing garment images were utilized to improve output quality and provide better visual appearance.

Comparison with Previous Task

In the previous task, the system mainly relied on basic body detection and simple cloth placement techniques. The garment overlay was static, meaning resizing and positioning had to be adjusted manually. Cloth fitting accuracy was moderate because the placement depended mostly on approximate body regions rather than accurate landmark mapping.

In today's implementation, the system became more intelligent and realistic. Dynamic garment scaling automatically adjusted clothing size according to body dimensions, improving overall fitting accuracy. Better shoulder alignment techniques reduced placement errors and ensured natural garment positioning. The use of transparent PNG garments and alpha blending significantly enhanced visual quality compared to the direct overlay approach used earlier.

Additionally, the earlier system used limited testing images and simple overlay logic, while today's implementation included multi-image testing and AI-based body understanding to determine proper overlay regions automatically. Overall, the updated system produced more realistic visualization, improved cloth fitting, and smoother integration between the user image and the virtual garment.

Improvements Achieved Today

- Improved realistic garment fitting
 - Enhanced shoulder alignment accuracy
 - Automatic dynamic garment resizing
 - Better cloth placement using pose estimation
 - Smooth cloth integration using alpha blending
 - Transparent PNG garment support
 - Intelligent overlay region detection
 - Improved body posture understanding
 - Enhanced visual quality and realism
 - Multi-image testing for better validation
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Working Principle

1. Capture or load the user image
 2. Detect body landmarks using pose estimation
 3. Identify shoulder and torso coordinates
 4. Resize garment dynamically according to body dimensions
 5. Align clothing based on body posture
 6. Apply cloth segmentation and overlay detection
 7. Blend garment using alpha blending techniques
 8. Generate realistic virtual try-on output
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Technologies Used

- Python

- OpenCV
- MediaPipe
- NumPy
- Pose Estimation
- Alpha Blending
- Cloth Segmentation

Observation

Compared to the previous implementation, today's output generated more accurate cloth fitting and improved garment positioning. Dynamic scaling allowed automatic garment resizing according to body proportions, while improved shoulder alignment reduced positioning errors. The use of transparent PNG clothing assets and alpha blending created a more natural and realistic virtual try-on appearance. Multi-image testing further confirmed that the updated implementation performs better under different user poses and body conditions.

Output

Input Person → Cloth Input → Advanced 2D Try-On Output



